ACCCALC DOCUMENTATION

# AccCalc v8.1.0

This handoff package documents the final `8.1.0` state of AccCalc after the focused follow-up release that fixed Windows local development, cleaned up the theme settings experience, and added transfer-pricing support to the managerial-accounting catalog.

## Release Summary

- Fixed the Vite config startup regression affecting `npm run dev` on Windows.
- Moved runtime Vite loading to `vite.config.mjs` with native loading.
- Made package metadata parsing BOM-safe inside the config path.
- Fixed theme-family picker wrapping and stale appearance summary behavior.
- Added `Transfer Pricing Support` with route, discovery, study, and workpaper wiring.

## Architecture Overview

### Routes

Route registration lives in `src/App.tsx` and points to feature pages in `src/features/...`.

### Shared Math

`src/utils/calculatorMath.ts` is the formula source of truth for reusable business and accounting logic.

### Solve Definitions

`src/utils/formulaSolveDefinitions.t` turns shared math into user-facing

### Discovery

`appCatalog`, `appSearch`, Smart Solver, and OCR routing work together

calculators with targets, warnings, and explanations.

so users can actually find the right tool.

## Important Files

- `src/utils/appCatalog.ts` : route metadata, aliases, categories, and release surfacing
- `src/features/smart/smartSolver.engine.ts` : Smart Solver field metadata and route scoring
- `src/features/smart/smartSolver.targets.ts` : explicit solve-target intent mapping
- `src/features/scan-check/services/ocr/ocrRouting.ts` : OCR route suggestion rules
- `src/utils/formulaStudyContent.tsx` : calculator-linked study panels
- `src/features/study/studyExpansion450.ts` : broader reviewer-topic expansion
- `src/features/workpapers/workpaperTemplates.ts` : workbook-template registry
- `src/utils/appSettings.ts` and `src/utils/themePreferences.ts` : persistent theme and preference state
- `src/utils/appRelease.ts` : current runtime release notes shown inside the app

## Vite And Local Development Fix

The highest-priority `8.1.0` fix was restoring local development on Windows.

> Root cause: the previous startup path allowed Vite config loading to go through a fragile bundled path that choked on Tailwind's native Windows dependency. The fix was to move runtime loading to `vite.config.mjs` and run it with `--configLoader native` .

The config now reads `package.json` with BOM-safe parsing before injecting the runtime version string. `vite.config.d.ts` is only a TypeScript-side stub to keep config tooling explicit without reintroducing the old runtime problem.

## Theme System And Settings Cleanup

Theme state has two axes: mode ( `system` , `light` , `dark` ) and family ( `classic` , `ocean` , `slate` , `rose` , `blossom` , `lavender` , `sunset` , `emerald` ).

- Theme state is restored before paint through `index.html` .
- The shell applies `data-theme` and `data-theme-family` for CSS-token theming.
- `8.1.0` fixes the larger-screen theme-card overlap by using a more forgiving grid and wrapped swatches.
- The quick appearance summary now updates when the theme family changes.

## New Academic Addition

### Transfer Pricing Support

The new managerial route helps students and reviewers read minimum transfer price, outside-market ceiling, and negotiation-range width without hiding the underlying relevant-costing logic.

- Route page: `src/features/business/TransferPricingSupportPage.tsx`
- Shared math: `computeTransferPricingSupport` in `calculatorMath.ts`
- Solve definition: `transferPricingSupportSolveDefinition`
- Study topic and formula support wired through the study system
- Workpaper template: `transfer-pricing-support`

## Testing And Validation

The final `8.1.0` state was validated with:

```
npm test
npm run build
```

The regression suite now includes transfer-pricing math, solve-definition alignment, route search coverage, Smart Solver target intent, and workpaper template registration.

## Next Safe Improvement Targets

- Reduce the chunk sizes for `WorkpaperStudioPage`, `appCatalog`, `formulaSolveDefinitions`, and `main`.
- Continue grouping and surfacing the larger catalog with more guided category hubs.
- Keep new curriculum additions attached to shared math, study content, discovery, and workpapers together.